

Pharma Agentic AI – Analysis Report

Automated opportunity assessment powered by Agentic AI (Master + Worker agents).

Query	Evaluate MOL2 in the Indian Chronic Kidney Disease space, highlighting decline risk, unmet needs and whitespace.		
Molecule	MOL2	Therapy area	Chronic Kidney Disease
Region / Country	India	Agents run	MARKET PATENT TRIALS EXIM INTERNAL GUIDELINES
Generated on	13 Dec 2025, 11:20 PM		

Executive Summary

- Market & growth view: Total market size: 10589.9 Mn USD with average 5Y CAGR 6.2%. Records: 19.
- Supply and trade dynamics: Total imports: 24269.6 tonnes, total exports: 26074.8 tonnes across 260 records.
- IP barriers and freedom-to-operate: Found 156 relevant patents. Earliest expiry year: 2003.
- Clinical development and pipeline: Found 16 clinical trials matching the criteria.
- Internal strategy and unmet needs: Evaluating MOL2 in Indian Chronic Kidney Disease (CKD) highlights potential repurposing opportunities and unmet needs, but also reveals operational constraints.

Agent-level Insights

IQVIA Insights Agent
High-level summary
Total market size: 10589.9 Mn USD with average 5Y CAGR 6.2%. Records: 19.
Structured details

Table:

- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2019 **sales_usd_mn:** 316.75 **cagr_5y:** -4.85 **competitors:** 22 **patient_burden_index:** 0.26
- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2024 **sales_usd_mn:** 861.42 **cagr_5y:** 15.4 **competitors:** 16 **patient_burden_index:** 0.68
- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2022 **sales_usd_mn:** 976.84 **cagr_5y:** 1.24 **competitors:** 22 **patient_burden_index:** 0.97
- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2021 **sales_usd_mn:** 497.54 **cagr_5y:** 8.97 **competitors:** 14 **patient_burden_index:** 0.95
- **molecule:** MOL2 **indication:** Chronic Kidney Disease **country:** India **year:** 2017 **sales_usd_mn:** 480.25 **cagr_5y:** 8.41 **competitors:** 22 **patient_burden_index:** 0.73
- ... (5 more records)

EXIM Trade Agent

High-level summary

Total imports: 24269.6 tonnes, total exports: 26074.8 tonnes across 260 records.

Structured details

Table:

- **molecule:** MOL2 **formulation:** Tablet **country:** India **year:** 2015 **import_tonnes:** 154.23 **export_tonnes:** 104.66
- **molecule:** MOL2 **formulation:** Patch **country:** India **year:** 2015 **import_tonnes:** 64.44 **export_tonnes:** 184.9
- **molecule:** MOL2 **formulation:** Capsule **country:** India **year:** 2019 **import_tonnes:** 19.83 **export_tonnes:** 81.0
- **molecule:** MOL2 **formulation:** Ointment **country:** India **year:** 2021 **import_tonnes:** 2.47 **export_tonnes:** 121.37

- **molecule:** MOL2 **formulation:** API **country:** India **year:** 2017 **import_tonnes:** 110.77 **export_tonnes:** 87.92
- ... (5 more records)

Patent Landscape Agent

High-level summary

Found 156 relevant patents. Earliest expiry year: 2003.

Structured details

Patents:

- **patent_id:** US1000426 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** GlobalBio **filing_year:** 2011 **expiry_year:** 2031 **status:** Granted **fto_flag:** Medium
- **patent_id:** US1001908 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** ABC Pharma **filing_year:** 2005 **expiry_year:** 2021 **status:** Pending **fto_flag:** High
- **patent_id:** US1003136 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** NovaCure **filing_year:** 2005 **expiry_year:** 2027 **status:** Pending **fto_flag:** Medium
- **patent_id:** US1003185 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** ABC Pharma **filing_year:** 2018 **expiry_year:** 2031 **status:** Expired **fto_flag:** Low
- **patent_id:** US1003444 **molecule:** MOL2 **indication:** Chronic Kidney Disease **assignee:** ABC Pharma **filing_year:** 2017 **expiry_year:** 2029 **status:** Expired **fto_flag:** Low
- ... (5 more records)

Clinical Trials Agent

High-level summary

Found 16 clinical trials matching the criteria.

Structured details

Table:

- **trial_id:** CT-005276 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase I **status:** Recruiting **sponsor:** ABC Pharma **country:** India
- **trial_id:** CT-018940 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase I **status:** Suspended **sponsor:** ABC Pharma **country:** India
- **trial_id:** CT-019942 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase II **status:** Active **sponsor:** PrimeCare **country:** India
- **trial_id:** CT-022598 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase I **status:** Completed **sponsor:** PrimeCare **country:** India
- **trial_id:** CT-030017 **molecule:** MOL2 **indication:** Chronic Kidney Disease **phase:** Phase IV **status:** Terminated **sponsor:** XYZ Pharma **country:** India
- ... (5 more records)

Internal Knowledge Agent

High-level summary

Evaluating MOL2 in Indian Chronic Kidney Disease (CKD) highlights potential repurposing opportunities and unmet needs, but also reveals operational constraints.

Structured details

Documents:

- asthma_strategy_2024.pdf
- respiratory_field_insights.pdf

Key Opportunities:

- Targeting patients with steroid intolerance in CKD, who suffer from limited alternative therapies

- Developing low-dose formulations for pediatric CKD patients, addressing a currently unmet need
- Exploring repurposing MOL2 in CKD to leverage its existing safety profile and familiarity in a new therapeutic area
- Improving inhaler affordability and accessibility in Tier-2/3 cities, addressing a significant pain point for CKD patients

Key Risks:

- Lack of evidence on MOL2's efficacy and safety in CKD, highlighting the need for extensive clinical trials
- Competing with established treatments in the CKD market, which may hinder market adoption
- Operational challenges in expanding distribution and logistics to reach rural hospitals and patients in Tier-2/3 cities

This report is generated by the Pharma Agentic AI prototype for EY Hackathon 6.0. Insights are based on simulated / mock data sources and should not be used as a substitute for validated market research.